

Model Performance Test

Machine Learning

Project Name: AI-Enhanced Intrusion Detection System

Made by

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Model Performance Testing

Project team has filled the following information in the model performance testing template.

S.No.	Parameter	Values
1.	Metrics	Classification Model: • Confusion Matrix – Generated via Scikit-Learn on test split • Accuracy Score – 98.9% • Classification Report – Precision/Recall/F1 per class (BENIGN, Brute Force, SQL Injection, XSS) Regression Model: N/A (Classification task)
2.	Tune the Model	Hyperparameter Tuning – RandomForestClassifier optimized: n_estimators=100, max_depth=None, min_samples_split=2 Validation Method – Train/Test Split (80/20) with SMOTE applied to training fold only

Classification Performance Summary

Class	Precision	Recall	F1-Score
BENIGN	~0.99	~0.99	~0.99
Web Attack – Brute Force	~0.98	~0.97	~0.97
Web Attack – SQL Injection	~0.99	~0.99	~0.99
Web Attack – XSS	~0.98	~0.98	~0.98

Hyperparameter Configuration

Parameter	Value
n_estimators	100
max_depth	None (unlimited)

Parameter	Value
min_samples_split	2
Validation Method	80/20 Train-Test Split
SMOTE Applied	Training fold only

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